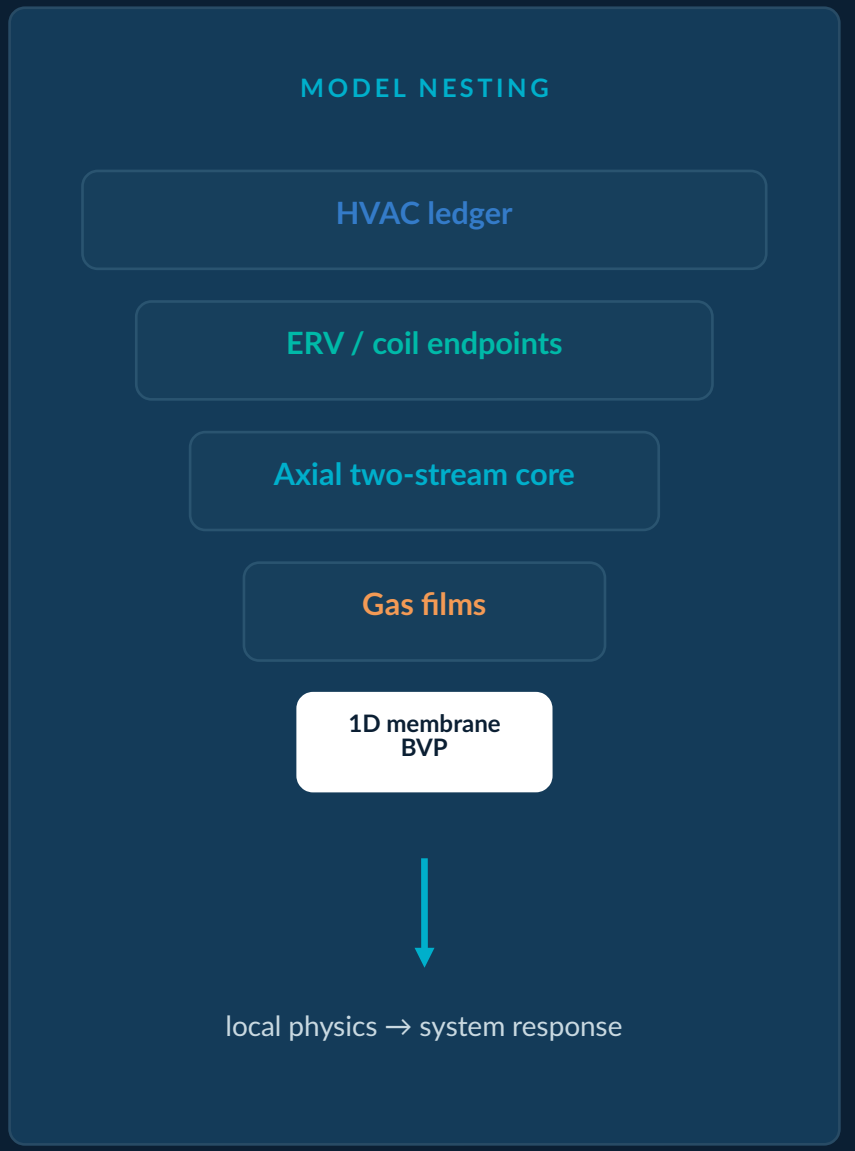
HEIQ / XMEMBRANE

Active-ERV Model

Technical Formulation and Numerical Implementation

Peer-review deck

Control volumes • constitutive laws • boundary conditions • solver hierarchy • conservation audits • validation map



Model basis: controlled repository state, 02 September 2026

REVIEW SCOPE

What is being reviewed—and what is not

The review target is the model formulation and implementation, not the commercial viability claim.

Reviewed model object

Steady moist-air states coupled to voltage-controlled water transport through an active membrane, integrated across an ERV core and linked to a reduced cooling-coil model.

Primary review questions

Are the governing equations internally consistent?
Are boundary conditions physically admissible?
Are numerical roots and balances closed?
Are assumptions visible?

Excluded from proof

Commercial membrane availability, long-term durability, complete receiver hardware, full thermal management, CAPEX and annual building performance.

MODEL OUTPUTS

- LOCAL J AND NW
- SURFACE ACTIVITIES
- REQUIRED AREA
- WATER BALANCE
- EO POWER
- REMAINING COIL POWER

Review principle: distinguish numerical verification, experimental validation and system closure.

Architecture and nested control volumes

CV-1 Outdoor / return air
bulk psychrometric states

→

CV-2 Active ERV core
heat + passive moisture + EO

→

CV-3 Cooling coil
ADP, condensate, reheat

→

CV-4 Power ledger
compressor + fans + EO + losses

CV-2 is itself nested

gas film → membrane x ∈ [0,L] → gas film

Conservation contract

water removed from process
= water gained by receiver

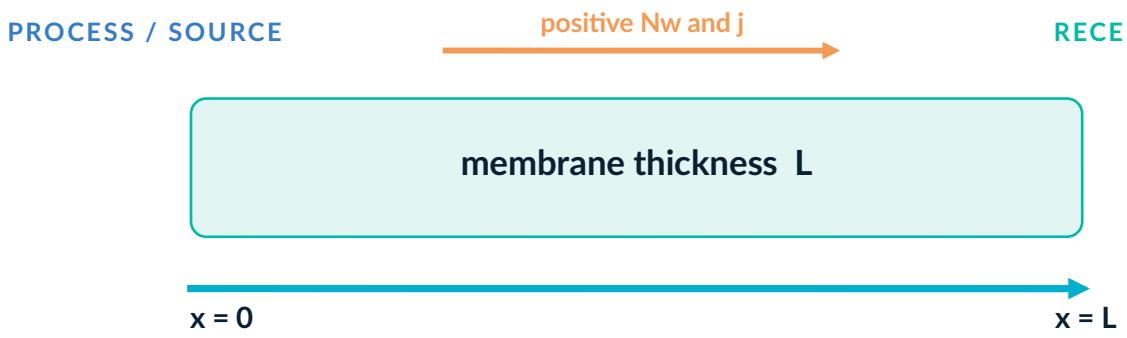
Architecture used in the selected interpretation: EO embedded in the ERV membrane; no separate post-ERV EO stage.

Canonical numerical chain and source ownership

1	**Local membrane transport (V2)**	1D activity + voltage; analytical power-law or H-Nafion BVP	`heiq_pre_receiver_material_analyses/` `heiq_pre_receiver_model.py`
2	**Axial process integration (V3)**	local state re-solved as process humidity changes with area	`src/heiq/wp1/axial/*`
3	**Coupled gas receiver (V12.1)**	process and receiver updated with the same transported water	`src/heiq/wp1/receiver/coupled_integration.py`
4	**Direct active-ERV screening**	balanced endpoints + segmented local sizing + coil linkage	`src/heiq/wp1/studies/direct_eo_erv.py`
5	**Study / optimisation workflows**	targets, parameter sweeps, basis checks and result auditing	`src/heiq/wp1/studies/* and scripts/*`

Controlled model map: docs/MODEL_CHAIN.md. Research models; generated outputs should be reproducible from source.

Coordinates, signs and conserved quantities



Primary conserved rate

Steady local water flux Nw [mol m^{-2} s^{-1}].

Integrated transfer is obtained from ∫A Nw dA and must match both air-stream balances.

Humidity ratio

w = kg water / kg dry air

Water activity

aw = pv / psat(T)

Electrical sign

positive j follows source → receiver

Thermal basis

steady bulk states; no dynamic storage

Positive-flux convention is declared in heiq_pre_receiver_model.py and used by the axial/receiver APIs.

Moist-air states are calculated on a dry-air basis

$$\mathrm{psat}(T) = \text{Buck equation over liquid water}$$

$$w = 0.621945 \cdot pv \,/\, (p - pv)$$

$$aw = pv \,/\, \mathrm{psat}(T)$$

$$h = 1.006T + w(2501 + 1.86T)$$

$$v = Rd\ T(1 + 1.6078w) \,/\, p$$

STATE VECTOR

$$\{\ T,\ RH,\ w,\ h,\ Tdew,\ v,\ p\ \}$$

- RH and activity are numerically identical for ideal moist air at the same temperature basis.
- Humidity ratio, not RH, is the conserved air-stream water coordinate.
- The model rejects vapor pressure ≥ total pressure and negative humidity ratio.

AIRFLOW BASIS

actual inlet CFM

dry-air mass flow derived from dry-air partial pressure

or standard dry-air CFM

reference T and p explicitly declared

The basis must not change silently between cases.

Implementation: src/heiq/wp1/airside/properties.py. Saturation pressure: Buck correlation.

ERV endpoints are effectiveness-based and balance-closed

$$T_{OA,out} = T_{OA,in} - \varepsilon_{s,eff}(T_{OA,in} - T_{exh,in})$$

$$w_{OA,out} = w_{OA,in} - \varepsilon_{l,eff}(w_{OA,in} - w_{exh,in})$$

$$\varepsilon_{eff} = \varepsilon \cdot \min(1, \dot{m}_{exh} / \dot{m}_{OA})$$

Endpoint closure

Receiver outlet humidity is obtained from the water balance. Receiver outlet enthalpy is obtained from the total moist-air enthalpy balance.

ACCEPTANCE CHECKS

- ✓ 0 ≤ sensible and latent effectiveness ≤ 1
- ✓ positive dry-air flows and fan efficiency
- ✓ water-balance residual finite and closed
- ✓ enthalpy-balance residual finite and closed
- ✓ no supersaturated outlet state
- ✓ both-side pressure-drop fan power included

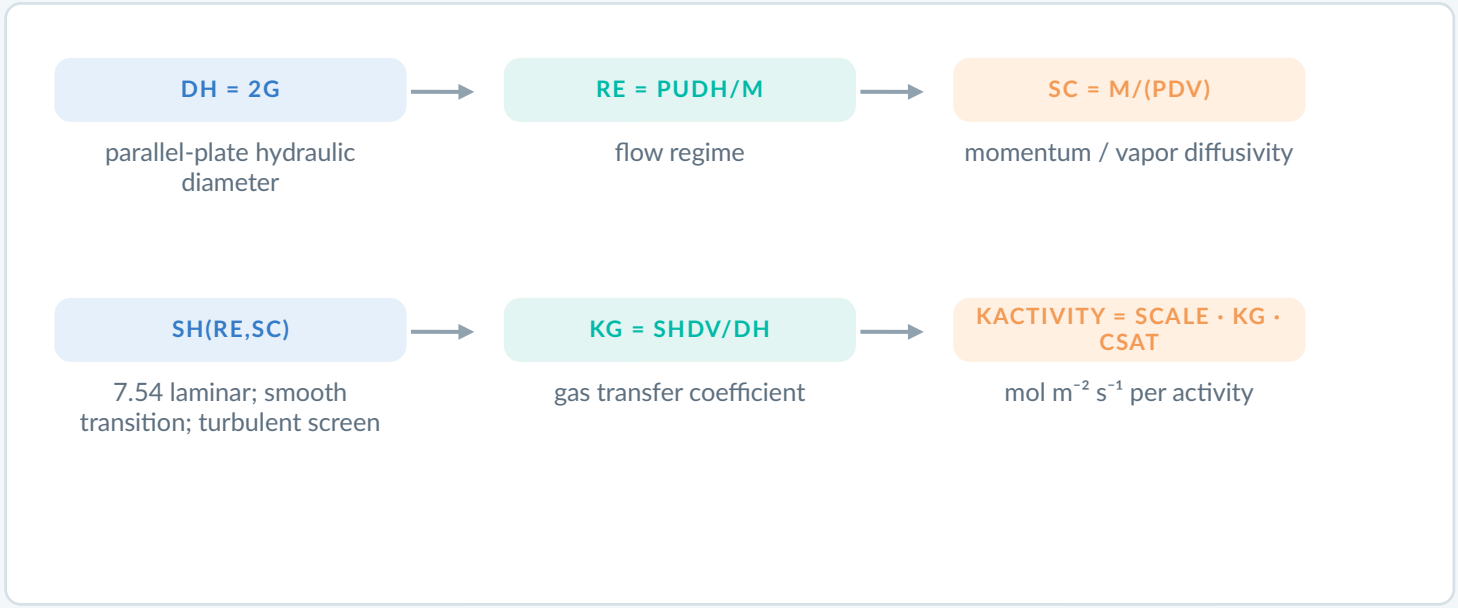
$$P_{fan} = [\dot{m}_{OA}\, v_{OA}\, \Delta p + \dot{m}_{exh}\, v_{exh}\, \Delta p] / \eta_{fan}$$

Implementation: run_erv() in src/heiq/wp1/studies/direct_eo_erv.py.

Finite gas films connect bulk air to membrane surface activity

$$\mathrm{Nw} = \mathrm{Ks}(\mathrm{as,bulk} - \mathrm{as,surface})$$

$$\mathrm{Nw} = \mathrm{Kr}(\mathrm{ar,surface} - \mathrm{ar,bulk})$$



Review implication

A high intrinsic membrane flux does not guarantee a high system flux. Film coefficients depend on temperature, channel gap, velocity, pressure and the selected correlation regime.

Implementation: calculate_film_state() in src/heiq/wp1/airside/properties.py.

Steady water flux combines EO drag and Fickian back-diffusion

$$\mathrm{Nw} = \mathrm{nd}(\mathrm{aw}) \cdot \mathrm{j}/\mathrm{F} - \mathrm{Dw}(\mathrm{aw})\ \mathrm{Sw}(\mathrm{aw}) \cdot \mathrm{daw}/\mathrm{dx}$$

EO contribution

NEO = nd j/F

nd (or tw) is mol H_2O transported per mol elementary charge.

Diffusive contribution

Ndiff = −Dw Sw daw/dx

The sign follows the local activity gradient and can oppose EO.

Steady-state constraint

Nw is spatially constant through a 1D membrane with no internal water source, sink or accumulation.

positive transport: process → receiver

Model class: one-dimensional, steady, isothermal within each local solve; local properties vary with water activity.

Governing equations declared in heiq_pre_receiver_material_analyses/heiq_pre_receiver_model.py.

Material properties are local functions of water activity

$$D(aw) = Dref\ (aw/aref)\hat{}pD$$

$$\sigma(aw) = \sigma ref\ (aw/aref)\hat{}p\sigma$$

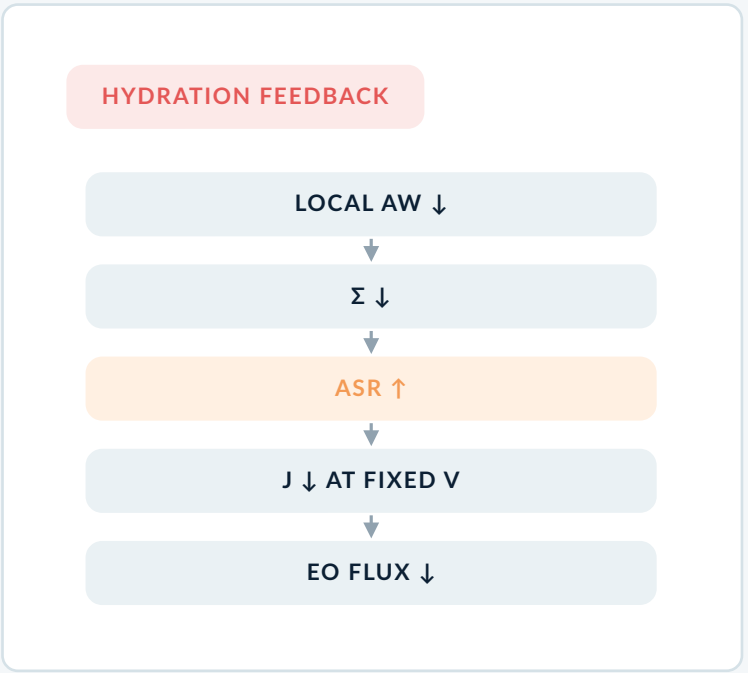
$$S(aw) = Sref\ (aw/aref)\hat{}pS$$

$$B(aw) = D(aw)S(aw)$$

INVERSE-DESIGN VECTOR

$$\{\ nd,\ Dref,\ \sigma ref,\ Sref,\ L,\ pD,\ p\sigma,\ pS,\ ASRseries\ \}$$

A feasible point is a simultaneous property combination—not evidence that one real material exhibits the full vector.



nd is an input for the material-neutral power-law model; it is not derived from D, S or σ.

The alternative H-Nafion implementation derives local hydration-dependent D, σ, S and drag from declared correlations.

The voltage equation contains ohmic and water-activity terms

$$dV/dx = j/\sigma(aw) + nd(aw)\ RT/F \cdot d\ ln(aw)/dx$$

Membrane ohmic

Vohm = $j \int_0^L dx/\sigma(aw)$

hydration-dependent ASR

Water polarization

Vwater = $\int nd\ RT/F \cdot d\ ln\ aw$

reversible activity work

Series resistance

Vseries = $j \cdot ASR_{series}$

contacts and current collectors

Fixed voltage

Vfixed

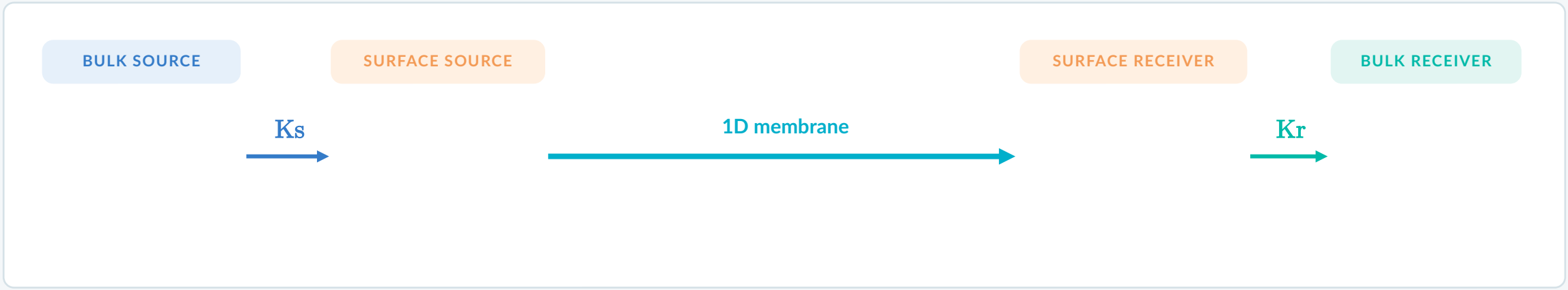
electrode / declared lower-bound loss

$$V_{terminal} = V_{ohm} + V_{water} + V_{series} + V_{fixed}$$

Applied terminal voltage is the control input; current density is the solved response.

Power accounting uses applied terminal voltage, not a nominal voltage cap.

Gas-film equations determine the membrane surface activities



$$a_{s,surf} = a_{s,bulk} - N_w/K_s$$

$$a_{r,surf} = a_{r,bulk} + N_w/K_r$$

$$0 < a_{s,surf},\ a_{r,surf} < 1$$

Physical flux bracket

Finite films impose row-specific lower and upper flux bounds before a surface reaches aw = 0 or 1.

Surface-direction audit

Midpoint and n+1 segment-boundary states are checked so a limiting end state is not hidden.

Activity bounds use ACTIVITY_EPS = 1×10^{-7}. Surface violations fail the local solution.

The power-law model reduces to a monotone scalar flux equation

$$R(Nw) = Nw - nd\ j/F + (Bref/L) \int as,surf\hat{}ar,surf\ (a/aref)\hat{}(pD+pS)\ da = 0$$

EXACT POWER-LAW REDUCTION

```
1  derive admissible flux interval from films
2  initialize with mean-property conductance
3  update surface activities from Nw
4  evaluate exact power-law integral
5  Newton update Nw within physical bracket
6  reconstruct ASR and voltage components
7  accept only if residual and surfaces pass
```

WHY THIS IS USEFUL

- Constant drag removes the need for a spatial BVP solve.
- D, S and σ still vary locally through analytical power-law integrals.
- The flux residual tolerance is evaluated explicitly.
- The activity profile can be reconstructed exactly after convergence.

$$ASRmem = L \cdot [\int a\hat{}(pD+pS-p\sigma)\ da] \ / \ [\sigma ref \int a\hat{}(pD+pS)\ da]$$

Default screening configuration: 12 Newton flux iterations; relative flux residual tolerance 1×10^{-7}.

The H-Nafion branch solves activity and voltage as a nonlinear BVP

UNKNOWN FIELDS AND PARAMETERS

$$y(\xi) = [\ aw(\xi),\ V(\xi)\ ]$$

$$p = [\ Nw,\ j\ ]$$

$$daw/d\xi = L[nd\ j/F - Nw]/[D\ S]$$

$$dV/d\xi = L\ j/\sigma + nd\ RT/F \cdot (1/aw)daw/d\xi$$

$$\xi = x/L \in [0,1]$$

FOUR BOUNDARY CONDITIONS

$$aw(0) = as,bulk - Nw/Ks$$

$$aw(1) = ar,bulk + Nw/Kr$$

$$V(0) = 0$$

$$V(1) = Vapplied - Vfixed - jASRseries$$

Local λ(aw), D(λ,T), σ(λ,T), S(aw) and nd(λ) are evaluated inside the ODE.

SciPy solve_bvp; default tolerance 2×10^{-5} and maximum 400 mesh nodes. Correlation-domain validity is reported separately.

Current is obtained by a bracketed root solve at each local state

VOLTAGE → CURRENT

```
given Vapplied

1  set jlow = 0
2  derive jhigh from gas-film surface limit
3  solve local flux/profile at jlow and jhigh
4  verify Vapplied is bracketed
5  bisect current interval
6  re-solve local state at jmid
7  stop when |Vcalc − Vapplied| ≤ tolerance
```

DEFAULT NUMERICAL CONTROLS

Bisection iterations	**36**
Absolute voltage tolerance	**2×10^{-6} V**
Relative voltage tolerance	**2×10^{-6}**
Internal search ceiling	**numerical guard**

INTERPRETATION

The internal current ceiling bounds the root search. It is not an engineering acceptance limit.

A legacy 21 A/m² acceptance gate has been superseded. Current density is reported as a diagnostic unless an explicit sensitivity is declared.

Voltage root implementation: solve_powerlaw_voltage_control().

Local states are rejected with explicit failure categories

REQUIRED FOR ACCEPTANCE

- ✓ voltage root bracket exists
- ✓ profile / scalar solve converged
- ✓ 0 < surface activities < 1
- ✓ finite positive current
- ✓ finite positive net flux
- ✓ terminal-voltage reconstruction passes

REPORTED FAILURE MODES

- ! below reversible voltage
- ! physical/profile bracket failure
- ! voltage root non-convergence
- ! activity profile out of bounds
- ! non-positive water flux
- ! internal search ceiling encountered

Missing or failed components remain unavailable; they are not silently set to zero.

Failure mapping: status_category() and coupled_local_gate().

The local membrane problem is re-solved as the air state changes with area

$$\mathrm{dwprocess/dA} = -\mathrm{Mw\ Nw\ /\ \dot{m}da,process}$$

$$\mathrm{dTprocess/dA} = \mathrm{fheat(Vj)\ /\ [\dot{m}da\ cp,moist]}$$

Thermal modes

Isothermal: dT/dA = 0.
Adiabatic upper bound: a declared fraction of local cell power heats the process air.

PREDICTOR–CORRECTOR

```
for each area step ΔA:
  solve local state at step start
  predict w and T at step end
  solve local state at predicted end
  halve ΔA if end state fails
  integrate mean flux and mean power
  shorten final step exactly at target
  update air state and continue
```

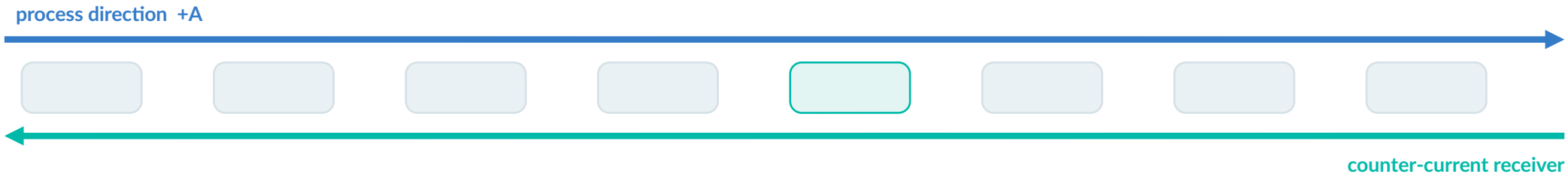
AUDITS

- ✓ target event resolved
- ✓ water integral = air balance
- ✓ power integrated by area
- ✓ max j / min Nw retained
- ✓ step failure categorized

Why area is the integration coordinate — The duty is proportional to local flux × active area, and the required total area is an output.

Native derivative and local-state interfaces: src/heiq/wp1/axial/local_state.py and axial/integration.py.

The same local flux updates process and receiver water balances



$$\Delta \dot{m}w = \bar{N} \quad w \; Mw \; \Delta A$$

$$wP,next = wP - \Delta \dot{m}w / \dot{m}P$$

$$wR,next = wR \pm \Delta \dot{m}w / \dot{m}R$$

Counter-current boundary construction

Integration starts at the process inlet paired with the receiver outlet. That receiver outlet is conditioned by the target and the global water balance.

Published only when closed

If the process target is not reached, the assumed counter-current receiver outlet is not a solved physical boundary and is returned as unavailable.

Required residuals: process balance + receiver balance + counter-current far-end boundary.

Implementation: integrate_coupled_receiver_case() in src/heiq/wp1/receiver/coupled_integration.py.

Balanced endpoints are interpolated, then each segment is locally sized

1. **Choose process outlet w**
   defines requested total water duty
2. **Close receiver endpoint**
   same transferred water; balanced dry-air flows
3. **Prescribe sensible endpoint**
   effectiveness-based process temperature
4. **Interpolate bulk states**
   typically 8 counter-current segments
5. **Solve local membrane**
   midpoints plus n+1 physical boundaries
6. **Size and integrate**
   $A_i = \dot{n}_{w,i} / N_{w,i}$; $P = \Sigma_{ji} A_i$

Reduced-order assumption Bulk T and w are interpolated between balanced endpoints; they are not obtained from a conjugate axial heat-and-mass PDE.

Boundary robustness n segments require n+1 boundary solves so midpoint-only feasibility cannot hide an end-state reversal.

Implementation: build_direct_eo_endpoints(), build_segments() and solve_segmented_transport().

Two integration paths answer different questions

PATH A • COUPLED V12.1

State-marching receiver model

- Process and receiver humidity evolve step by step with area.
- Local membrane problem is re-solved at start and end of each step.
- Counter-current outlet is conditioned and audited at the far end.
- Adaptive step halving and exact target-event shortening.
- Best for target-reaching area / receiver sensitivity studies.

PATH B • DIRECT ACTIVE ERV

Endpoint-segmented screening model

- Balanced process and receiver endpoints are constructed first.
- Bulk states are linearly interpolated through a fixed segment count.
- Each segment is sized at a representative local state.
- Explicit boundary-state direction and voltage audits are added.
- Best for fast architecture / material frontier screening.

Outputs from the two paths are not interchangeable unless material, geometry, receiver and numerical basis are matched.

This distinction explains why different O30 area studies can produce different—but internally valid—numbers.

Remaining latent and sensible duty is solved from a reduced ADP model

$$\mathrm{wtarget} = \mathrm{BF} \cdot \mathrm{win} + (1 - \mathrm{BF}) \cdot \mathrm{wADP}$$

$$\mathrm{hoff} = \mathrm{BF} \cdot \mathrm{hin} + (1 - \mathrm{BF}) \cdot \mathrm{hADP}$$

$$\mathrm{Qcool} = \dot{m}\mathrm{da}(\mathrm{hin} - \mathrm{hoff})$$

$$\dot{m}\mathrm{cond} = \dot{m}\mathrm{da}(\mathrm{win} - \mathrm{woff})$$

$$\mathrm{Pcomp} = \mathrm{Qcool} \,/\, \mathrm{COP}$$

COP SCALING

$$\mathrm{COPCarnot} = \mathrm{Tevap} \,/\, (\mathrm{Tcond} - \mathrm{Tevap})$$

$$\mathrm{COP} = \mathrm{COPrated}\,[1 + \eta\mathrm{cons}(\mathrm{COPCarnot}/\mathrm{COPCarnot,rated} - 1)]$$

Physical checks — off-coil RH ≤ 1; supply RH ≤ 1; finite COP; reheat mode explicitly declared; “none” is invalid if reheat is required.

The coil is a reduced endpoint model—not a refrigerant-cycle simulation.

Implementation: solve_coil() and _scaled_cop() in src/heiq/wp1/studies/direct_eo_erv.py.

Membrane area is translated into channel length and fan power

Air volume flow
$\dot{V} = \dot{m}da \cdot v$

→

Total channel width
$Wtot = \dot{V}/(u\ g)$

→

Channel length
$\ell = A/(Nsides\ Wtot)$

→

Reynolds number
$Re = \rho u Dh/\mu$

→

Pressure drop
laminar or Blasius screen

→

Fan power
$Pfan = \dot{V}\Delta p/\eta fan$

$$\Delta p = 12\ \mu\ \ell\ u\ /\ g^2 \quad (Re \leq 2300)$$

$$\Delta p = f\ (\ell/Dh)\ \rho u^2/2, \quad f = 0.3164\ Re^{-0.25}$$

CONSERVATIVE SELECTION

Selected active-core fan power = max(native ideal-channel estimate, declared ERV pressure-drop floor).

This is a packaging screen; manifolds, maldistribution and fouling remain outside the model.

Implementation: src/heiq/wp1/engineering/channels.py and direct_eo_core_hydraulics().

Electrical identities are checked locally before system totals are formed

$$V_{terminal,j} = V_{membrane,j} + j^2 ASR_{series} + V_{fixed,j}$$

$$P_{EO,cell} = \int_A V_{terminal,j}\, dA$$

$$P_{EO,grid} = P_{EO,cell} / \eta_{electronics}$$

$$P_{known} = P_{AC} + P_{EO,grid} + P_{fans} + P_{common} + \text{included losses}$$

LEDGER DISCIPLINE

- ✓ component power identity checked per area step
- ✓ missing terms remain unavailable—not zero
- ✓ site / grid conversion efficiency declared
- ✓ common loads applied consistently to comparators
- ✓ system claim disabled if required component unresolved

Receiver regeneration and heat rejection are not closed in the selected gas-receiver architecture.

System savings are conditional until the full ledger is closed.

Boundary discipline is also documented in docs/SYSTEM_ENERGY_CLOSURE_STUDY.md.

The framework supports forward prediction and inverse design

FORWARD MODEL

measured material vector + operating point → system flux, area and power

INVERSE DESIGN

target humidity, voltage and area contract → required property region

DEFAULT SCREENING DOMAINS

nd	0.3–120	log	D_{ref}	10^{-12}–10^{-9} m²/s	log	σ_{ref}	0.1–50 S/m	log
L	10–100 µm	log / set	V	0.003–2.5 V	log	S_{ref}	10^{4}–10^{5} mol m^{-3} aw^{-1}	set
pD	0–4	uniform	$p\sigma$	0–8	uniform	pS	−1–2	uniform

Sampling is not a probability model unless distributions are physically justified.

Ranges shown from heiq_pre_receiver_material_analyses/config/default.json; downstream studies may override them.

Conservation, equivalence and convergence checks are explicit

41

selected relevant tests passed

Executed 02 Sep 2026

- ✓ membrane thermodynamics
- ✓ axial equivalence
- ✓ receiver API
- ✓ gradient closure
- ✓ direct active ERV
- ✓ O30 required-area workflow

RESIDUAL AUDITS

- ✓ local flux residual
- ✓ terminal-voltage error
- ✓ process water balance
- ✓ receiver water balance
- ✓ counter-current boundary
- ✓ segment continuity
- ✓ electrical component identity

STEP-HALVING AUDIT

20 m² step	**10,499.7174 m²**
10 m² step	**10,499.6770 m²**

relative difference **0.000385%**

Fixed case-94 basis at receiver multiplier 3.0.

Verification ≠ experimental validation

Test command covered six relevant test modules; convergence example from docs/O30_REQUIRED_AREA_1V_v1.md.

Published O30 numbers must be tied to their exact model contract

Study basis	Integration method	Fixed basis	Illustrative O30 area	Interpretation
Selected final-system interpretation	8-segment endpoint sizing	selected material frontier; architecture-specific endpoint assumptions	**5,899 m² at 1 V**	Used for the earlier 418.4 kW system interpretation
Nonlinear required-area study	coupled V12.1 target-reaching integration	legacy case-94 material + fixed channel geometry + receiver multiplier	**10,499.7–12,139.9 m² at 1 V**	Receiver airflow changes area and core power
Canonical local hierarchy	V2 local → V3 axial → V12.1 receiver	declared per workflow	**not one universal value**	Model chain; outputs depend on explicit contract

Different areas are not a contradiction when material, geometry, receiver boundary and integration method differ—but they must never be merged.

Sources: selected v9 interpretation and docs/O30_REQUIRED_AREA_1V_v1.md.

The model is strongest where conservation is explicit—and weakest where closure is prescribed

WELL CONSTRAINED NUMERICALLY

- ✓ local sign convention
- ✓ water balances
- ✓ voltage reconstruction
- ✓ surface admissibility
- ✓ target-event resolution

Verification evidence

PRESCRIBED MODEL CLOSURES

- ! sensible / latent effectiveness
- ! power-law property shapes
- ! film correlation and scale
- ! segment bulk interpolation
- ! scaled-Carnot COP

Sensitivity required

NOT YET EXPERIMENTALLY CLOSED

- ! vapor nd for target material
- ! RH-resolved ASR under load
- ! electrode heat and losses
- ! receiver water handling
- ! manifolds / fouling / lifetime

Experimental evidence required

The inverse-design property vector is a target region, not a validated material identification.

Each high-impact model term should have a direct measurement pathway

Model quantity	Experiment	Measured observable	Acceptance focus
Net EO transport Nw	matched polarity reversal + EO-off control	mass gain/loss on both gas streams	direction, repeatability, uncertainty
Effective drag nd	measure NEO and current at the same state	nd = NEO F/j	vapor-specific—not liquid-cell inferred
D(aw)S(aw)	passive activity-gradient series	steady diffusion flux vs. Δaw and L	separate films from membrane
σ(aw) / ASR	impedance or DC under representative RH	area resistance and temperature	same hydration as flux test
Gas films Ks, Kr	vary velocity and channel gap	flux response + blank-cell pressure drop	test Sh correlation and bypass
Voltage / heat ledger	simultaneous V, I and calorimetry	V components, electrical heat partition	stable operation near 1 V
Receiver closure	meter inlet/outlet humidity and any condensate	full donor/receiver water balance	saturation and removal mechanism

A coupon test should target the incremental EO duty—not the total active-core water transfer that already includes passive recovery.

PEER-REVIEW CONCLUSION

The model is auditable because its balances and failure modes are explicit.

STRENGTH

Conservation, voltage closure, surface bounds and numerical statuses are enforced rather than assumed.

KEY APPROXIMATION

The selected active-ERV path interpolates balanced bulk endpoints; it is not a conjugate axial PDE.

VALIDATION GAP

The inverse-designed material vector and receiver / thermal hardware remain experimentally open.

Recommended review focus Are the chosen constitutive ranges defensible? Is endpoint interpolation adequate? What data will close the receiver and heat ledger?